

The Three A's of Zero Trust

Attribution, Authority, and Authorization

Modern Zero Trust discussions often focus heavily on technologies.

Identity providers. Multi-factor authentication. Microsegmentation. Data tagging. Policy engines. Behavior analytics. Continuous monitoring.

All of those things matter.

But over time, I've found that most Zero Trust capabilities ultimately exist to support one or more of three fundamental concepts:

- Attribution
- Authority
- Authorization

The more I work through enterprise architectures, the more I believe these three concepts form the operational foundation beneath almost every Zero Trust activity, capability, and enforcement mechanism.

Not just technically – architecturally, philosophically, and operationally.

And importantly, they are not independent concepts. They recursively depend on one another.

Understanding those relationships changes how you think about Zero Trust itself.

Attribution

At its simplest, attribution answers a basic question, “who or what performed this action?”

But in mature Zero Trust architectures, attribution becomes far more expansive than simple user identification.

Attribution applies to:

- users
- devices
- workloads
- services
- applications
- APIs
- automation
- policies
- enforcement actions

- analytics outcomes
- orchestration actions
- telemetry sources
- data transformations
- administrative decisions

In other words, the architecture must be able to reliably associate actions, events, decisions, and state changes back to specific entities and systems.

This is foundational to trust.

Because without attribution:

- accountability collapses
- auditing weakens
- policy provenance disappears
- enforcement becomes ambiguous
- forensic reconstruction becomes unreliable
- automation becomes dangerous
- trust decisions lose confidence

And importantly, Zero Trust attribution is not only about identifying users.

It is also about attributing decisions themselves.

For example:

- Which system denied access?
- Which policy triggered enforcement?
- Which telemetry source elevated risk?
- Which orchestration engine isolated the device?
- Which authoritative source supplied the attribute?
- Which administrator approved the exception?
- Which enforcement point blocked the request?

These questions matter because enterprise trust cannot exist where decision lineage is unclear.

Attribution creates accountability.

But accountability alone is insufficient. Because attribution does not establish whether an entity or system should be trusted to act. That leads directly into authority.

Authority

Authority answers a different question, “who or what is trusted to define truth, make decisions, enforce policy, or establish state?”

This concept is frequently underappreciated in enterprise security architecture.

Many organizations discuss trust without clearly defining authority boundaries. But Zero Trust architectures fundamentally depend on authoritative sources. Without them, the system has no stable foundation for decision-making.

For example:

- What system is authoritative for identity?
- What system is authoritative for device posture?
- What system is authoritative for policy?
- What system is authoritative for data classification?
- What system is authoritative for risk scoring?
- What system is authoritative for access decisions?
- What system is authoritative for telemetry?
- What system is authoritative for enforcement?

These are not merely technical implementation questions. They are *trust-governance questions*. Because every authorization decision ultimately depends on authoritative truth sources.

If authoritative boundaries become ambiguous:

- policies conflict
- enforcement diverges
- telemetry loses confidence
- data integrity degrades
- trust chains weaken
- operators lose clarity
- automation becomes inconsistent

This problem appears constantly in enterprise environments.

Multiple systems begin independently:

- assigning permissions
- classifying risk
- making access decisions
- evaluating posture
- defining policy
- overriding enforcement
- managing identity state

Eventually, no one can confidently answer, “which system actually owns the decision?”

That ambiguity is dangerous.

Zero Trust architectures require clear authority boundaries because trust cannot be continuously evaluated against uncertain truth sources.

And authority itself is recursive.

For example:

- A policy engine must itself be trusted.
- An identity provider must itself be authoritative.
- Telemetry used for authorization must itself originate from trusted systems.
- Enforcement points must be trusted to correctly execute decisions.
- Analytics engines must be trusted to accurately interpret data.

Authority therefore becomes both an **operational** and **architectural** concern.

And critically, authority without attribution becomes unaccountable.

Which leads to the third concept.

Authorization

Authorization answers the runtime question, “should this specific action be permitted right now under these exact conditions?”

This is where Zero Trust fundamentally diverges from many traditional security models. Historically, organizations often treated authorization as relatively static. If identity was established and permissions existed, access was generally granted. Zero Trust changes this entirely.

Authorization becomes:

- contextual
- dynamic
- continuously reevaluated
- risk-informed
- policy-driven
- environment-aware

Authorization no longer depends solely on identity.

It depends on the evaluated trust state of:

- the user
- the device
- the session
- the network
- the data

- the environment
- the behavior
- the request pattern
- operational conditions
- active threat indicators
- policy constraints
- mission context

all evaluated together at runtime.

Importantly, authorization depends directly on both attribution and authority.

Authorization cannot function without reliable attribution because:

- unidentified actions cannot be evaluated
- unattributed telemetry cannot be trusted
- unauthenticated entities cannot be assessed

Authorization also cannot function without authoritative truth sources because:

- inaccurate identity state corrupts decisions
- stale posture data weakens trust evaluation
- conflicting policy sources create inconsistent enforcement
- unreliable analytics introduce decision instability

In practice, authorization becomes the operational convergence point of the entire architecture.

This is why modern Zero Trust environments place such heavy emphasis on:

- policy decision points
- contextual access evaluation
- runtime telemetry
- behavioral analytics
- continuous verification
- adaptive enforcement
- dynamic policy orchestration

Authorization is no longer a one-time event; it is a continuous operational process.

The Recursive Relationship

The Three A's are not isolated functions. They recursively support one another.

Authorization depends on authoritative sources.

Authority requires attribution.

Attribution depends on trusted authoritative telemetry.

Enforcement authority itself must be attributable.

Policy authorities must themselves be authorized.

Telemetry attribution depends on trusted collection mechanisms.

The relationships continually loop back into one another.

This is why Zero Trust architectures become difficult to implement at enterprise scale. The challenge is not simply deploying tools. The challenge is maintaining trustworthy relationships between:

- attribution chains
- authority boundaries
- authorization decisions

across thousands of systems, users, devices, policies, workflows, and enforcement mechanisms.

Most architectural failures occur when one of these chains becomes ambiguous.

For example:

- unclear policy ownership
- conflicting authoritative sources
- inconsistent enforcement
- unverifiable telemetry
- unattributed automation
- orphaned permissions
- disconnected analytics
- static authorization assumptions

These failures erode trust not because security controls are absent, but because trust relationships themselves become unreliable.

Zero Trust as a Trust Architecture

One of the biggest misconceptions about Zero Trust is the belief that it is primarily a collection of security technologies.

It is not.

At its core, Zero Trust is an architecture for establishing and continuously evaluating trust.

That architecture requires:

- reliable attribution
- authoritative truth sources
- contextual authorization

working together continuously.

Every capability ultimately contributes to one or more of those goals.

Identity systems. Policy engines. Analytics platforms. Data tagging. Device posture assessment. Behavioral monitoring. Access enforcement. Telemetry pipelines. Automation systems.

All of them support the establishment, validation, protection, or enforcement of attribution, authority, and authorization relationships across the enterprise.

The technologies matter, but the trust relationships matter more.

Final Thought

Attribution establishes accountability.

Authority establishes trusted truth.

Authorization establishes permissible action.

When those relationships are **clear**, trust decisions become **defensible**.

When those relationships become *ambiguous*, Zero Trust begins to *fail*.

The more I work through enterprise Zero Trust architectures, the more I believe the real challenge is not deploying security technologies. It is maintaining confidence in the attribution, authority, and authorization chains across the enterprise continuously, consistently, and at scale.

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